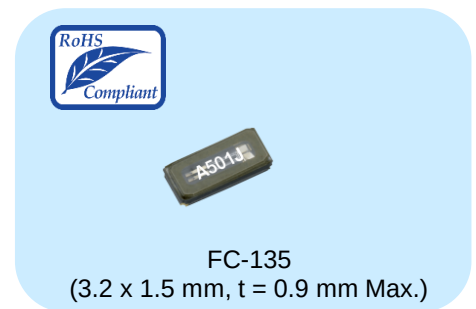


3.2 x 1.5 mm size 32.768 kHz Crystal Unit: FC-135

Features

- Package size: 3.2 x 1.5 mm, t = 0.9 mm Max.
- Nominal frequency range: 32.768 kHz
- Frequency tolerance: $\pm 20 \times 10^{-6}$ (+25 °C $\pm$ 5 °C)
- Operating temperature: -40 °C to +85 °C
(-40 °C to +105 °C) Please contact Epson
- ESR: 70 k Ω Typ.



Applications

- Wearable products
- Low power MCUs for sub-clock
- Wireless modules for sub-clock

Description

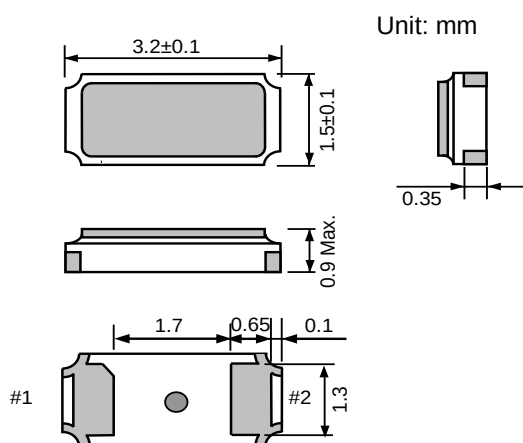
The FC-135 is a widely adopted 32.768 kHz crystal unit that has been used worldwide since its release in 2002.

Ideal for MCU sub-clocks and modules from consumer devices to industrial equipment applications.

For extended temperature range applications up to +105 °C, please contact Epson.

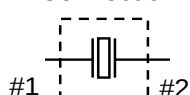
Epson is a leading supplier of kHz-band crystal units and offers a lineup of oscillators with built-in oscillator circuit ICs and real-time clock modules with built-in real-time clock ICs, in addition to crystal units.

Outline Drawing and Terminal Assignment



Unit: mm

Internal
Connection



Pin	Connection
#1	X'tal
#2	X'tal

[1] Product Number / Product Name

(1-1) Product Number

Q13FC1350xxxx00 (Please contact Epson for details)

(1-2) Product Name (Standard Form)

FC-135 32.768000kHz 12.5 +20.0-20.0

① ② ③ ④

①Model ②Frequency ③Load capacitance (pF) ④Frequency tolerance ($\times 10^{-6}$, +25 °C)

[2] Absolute Maximum Ratings

Item	Symbol	Rating value			Unit	Note
		Min.	Typ.	Max.		
Storage temperature range	T_stg	-55	-	+125	°C	
Maximum level of drive	GL	-	0.5	1.0	μW	Not destroyed

[3] Operating Conditions

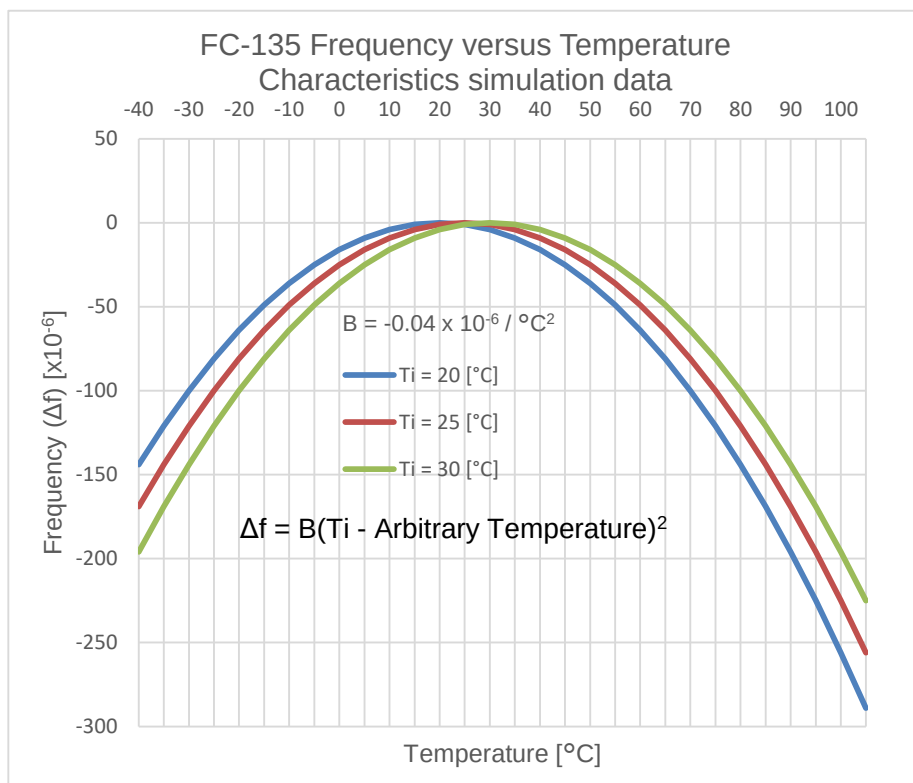
Item	Symbol	Rating value			Unit	Note
		Min.	Typ.	Max.		
Operating temperature range	T_use	-40	-	+85	°C	
		-40	-	+105		Please contact Epson
Level of drive	DL	0.01	0.1	0.5	μW	
		0.01	0.5	1.0		Please contact Epson

[4] Static Characteristics

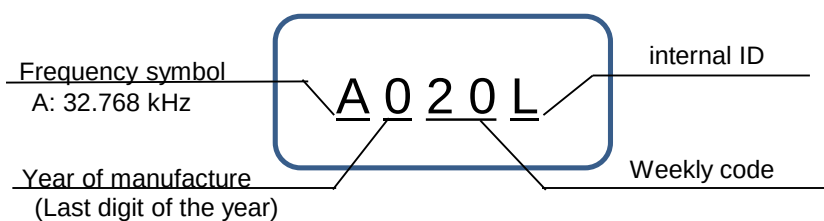
Item	Symbol	Specifications	Unit	Condition / Remarks
Nominal frequency range	f_nom	32.768	kHz	
Frequency tolerance	f_tol	±20	$\times 10^{-6}$	T_use = +25 °C ± 3 °C DL = 0.1 μW Does not include frequency aging
Turnover temperature	Ti	+25 ± 5	°C	
Parabolic coefficient	B	-0.04 Max.	$\times 10^{-6} / ^\circ\text{C}^2$	
Motional resistance (ESR)	R1	70 Typ.	kΩ	Measuring instrument: Saunders: 140-B CI meter DL = 0.5 μW
Motional capacitance	C1	3.4 Typ.	fF	
Shunt capacitance	C0	1.0 Typ.	pF	
Load capacitance	CL	7, 9, 12.5	pF	Please specify
Isolation resistance	IR	500 Min.	MΩ	
Frequency aging	f_age	±3 Max	$\times 10^{-6}$	T_use = +25 °C, DL = 0.1 μW First year

[5] Frequency versus Temperature Characteristics

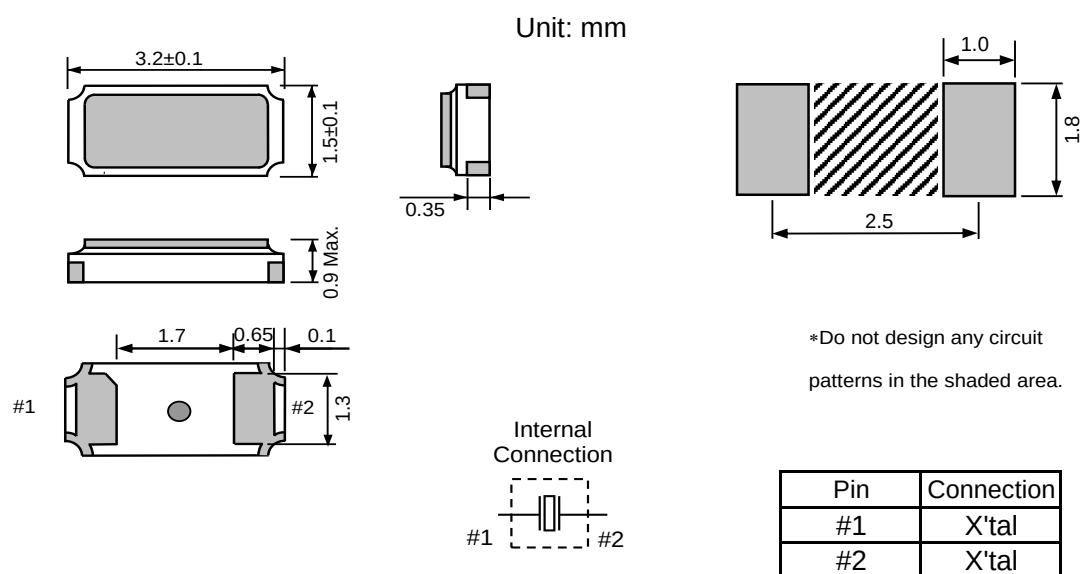
Simulation data



[6] Marking Description



[7] Outline Drawing and Recommended Footprint



Reference weight Typ.: 11 mg

Terminal coating: Au plating

[8] Moisture Sensitivity Level

Parameter	Specification	Conditions
MSL	LEVEL1	IPC/JEDEC J-STD-020D.01

[9] Reflow Profile

(1) Standard profile (Type. E)

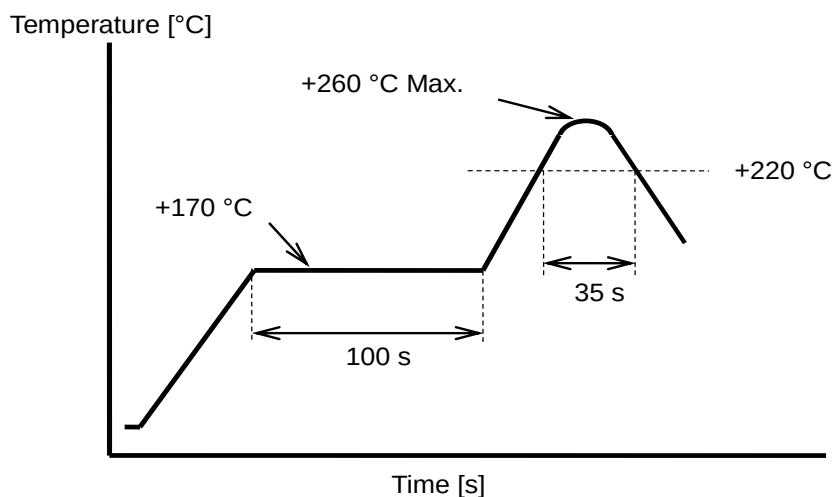
Conditions of hot air convection reflow

Pre heating temperature: +170 °C

Pre heating time: 100 s

Heating temperature: +220 °C

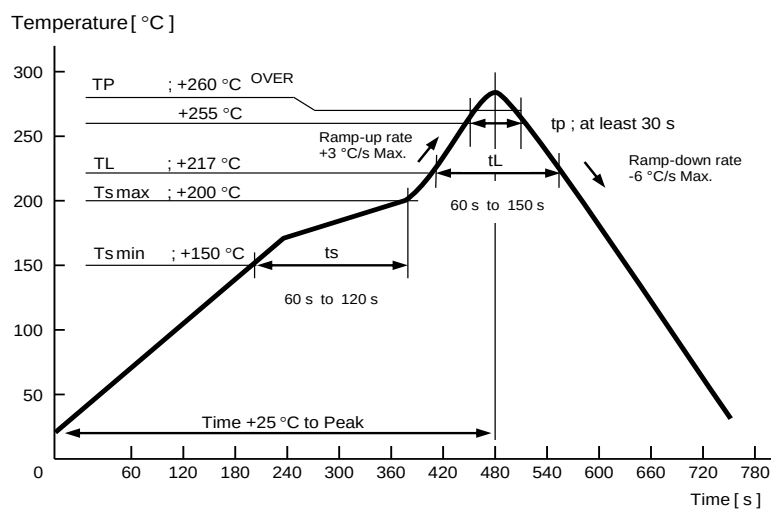
Heating time: 35 s



(2) Optional profile (IPC/JEDEC J-STD-020D.01)

Note: For the option, the shift amount in the reliability test is different.

Please contact Epson for details



[10] Packing Information

(1) Packing Quantity

The last two digits of the Product Number (Q13FC1350xxxx00) are a code that defines the packing quantity.
The standard is "00" for a 3 000 pcs/Reel.

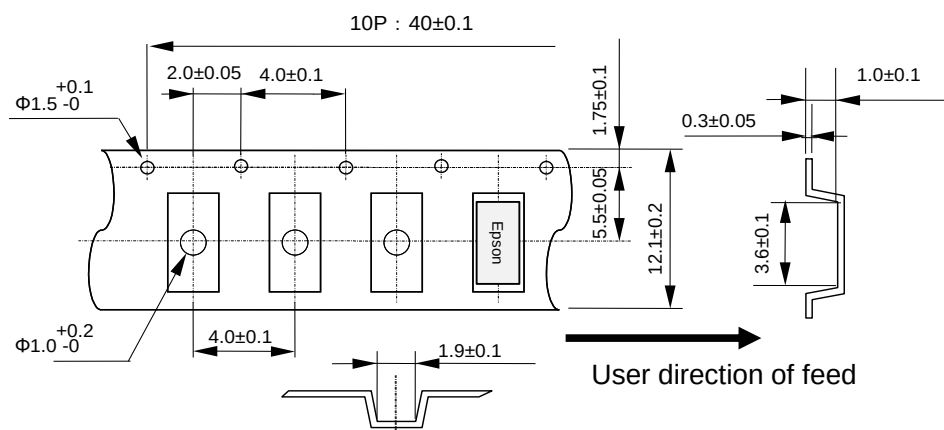
(2) Taping Specification

Compliant to EIA-481, IEC 60286 and JIS C0806

(2-1) Tape Dimensions

Carrier Tape Material : PS (Polystyrene)

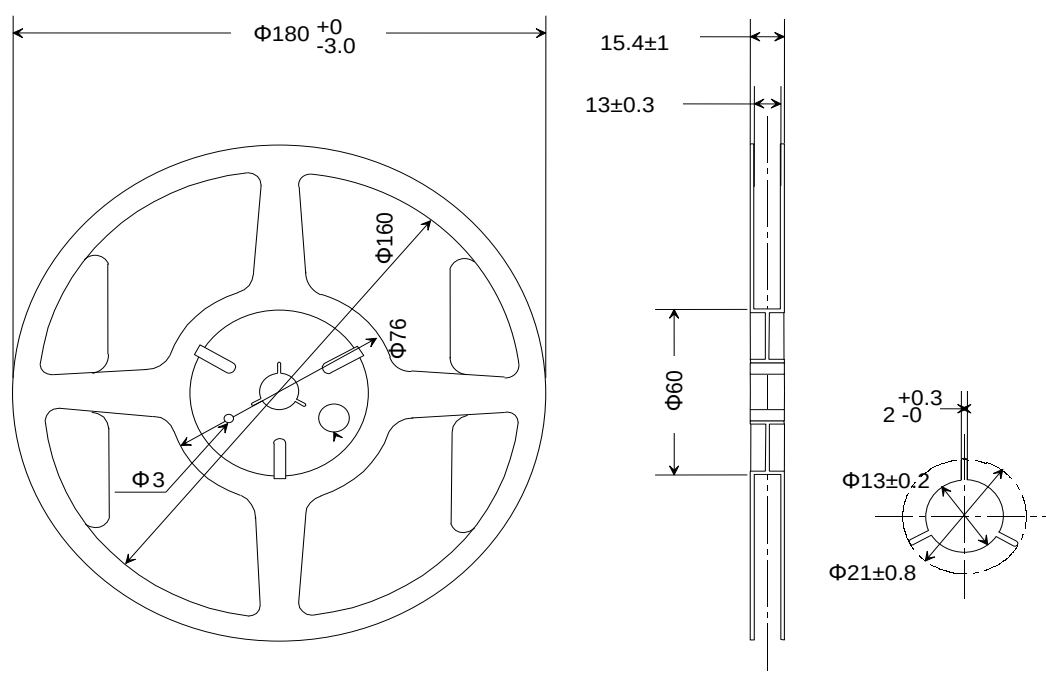
Top Tape Material : PET (Polyethylene Terephthalate) +PE (Polyethylene)



(2-2) Reel Dimensions

Center Material : PS (Polystyrene)

Reel Material : PS (Polystyrene)

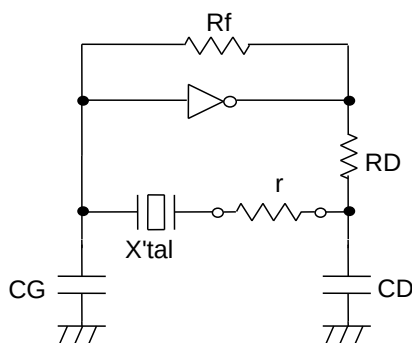


[11] Handling Precautions

Please review the "Handling Precautions" on our website for proper handling and behavior to ensure the performance of your equipment/product.
(<https://www5.epsondevice.com/en/information/#precaution>) In addition to the "Handling Precautions" on the website, please also pay attention to the following to avoid deterioration of product performance.

1. Max three (3) times reflow is allowed.
In case of rework by soldering iron, its condition should be +350 °C max. + within 5 sec.
2. Applying excessive shock or vibration to the crystal unit may causes deterioration damage.
The product may be damaged depends on the condition such as shock in assembly machine.
Please check if your condition is safe in advance.
And in case of assembly condition change, please check it again in advance.
3. Shortest line pattern on PCB is recommended.
Too long line on PCB may causes abnormal oscillation.
4. Failures covered by free warranty period are limited to the cases where the product is used under the usage and environment described in the specifications. In addition, products that have been opened (including partially opened, modified, or intended to be opened) are not covered. In order to ensure frequency accuracy and prevent moisture condensation due to sudden temperature changes, it is recommended to store and use in normal room temperature and humidity.
If the product is stored for a long period (one year or more), please check solderability of the terminals before use.
5. Ultrasonic cleaning may cause resonant damage of the crystal unit depend on its condition.
Since we are unable to specify the conditions (type of cleaning unit, power, time, condition inside the bath, etc.) at your company, we cannot guarantee the performance of the product when it is cleaned by ultrasonic cleaner.
6. Condensation on oscillator circuit board may causes frequency shift or oscillation stop.
Please use the product under the condition there is no condensation.
7. If excessive drive level is applied to the crystal unit, it may cause performance deterioration and damages. Please design appropriate drive level on the circuit.
8. Characteristics differences between our measurement and your company's measurement may occur depending on measurement method and conditions. Please check it thoroughly before use.
9. Do not place signal lines, power lines, or GND lines in mounting area of the product, its inner layer, or its back side. In order to avoid malfunction due to induction of other signal lines, please do not place signal lines near the product. It may affect product characteristics.
10. If there is no margin in negative resistance of the oscillator circuit, the crystal unit may not oscillate or may take a long time to oscillate. Therefore, negative resistance in the oscillator circuit should be at least five times of the crystal unit's equivalent series resistance. Please follow this circuit design rule.
11. Aging specifications are estimated value of frequency shift from reliability test results. It does not mean to guarantee product lifecycle.
12. If customer wants to use our product contrary to this caution and advice, please use it at your own risk.

<How to check the negative resistance>



- 1) Insert a pure resistance " r " in series with the crystal oscillator.
- 2) Adjust " r " to find the maximum " r " value that starts oscillation
- 3) Look at the value of " r " in the oscillation state of 2).

Negative resistance of the circuit $|-R| =$

$r + \text{crystal unit series resistance value } R1$

Guideline for negative resistance $|-R|$:

$|-R| > R1 \text{ Max. } \times 5$

PROMOTION OF ENVIRONMENTAL MANAGEMENT SYSTEM CONFORMING TO INTERNATIONAL STANDARDS

At Seiko Epson, all environmental initiatives operate under the Plan-Do-Check-Action (PDCA) cycle designed to achieve continuous improvements. The environmental management system (EMS) operates under the ISO 14001 environmental management standard.

All of our major manufacturing and non-manufacturing sites, in Japan and overseas, completed the acquisition of ISO 14001 certification.

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IATF 16949 is the international standard that added the sector-specific supplemental requirements for automotive industry based on ISO9001.

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●Complies with EU RoHS directive.

*About the products without the Pb-free mark.

Contains Pb in products exempted by EU RoHS directive

(Contains Pb in sealing glass, high melting temperature type solder or other)

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